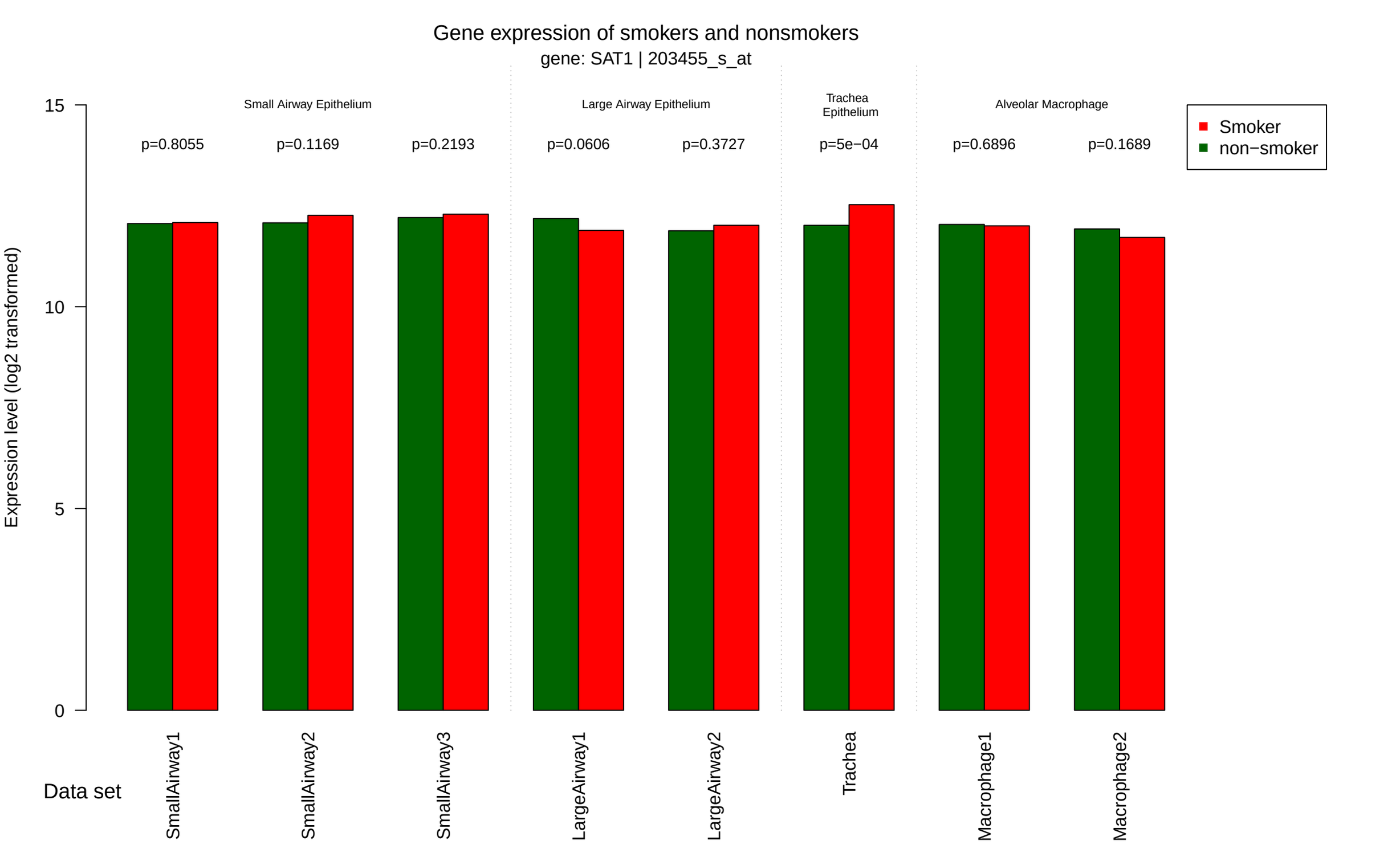
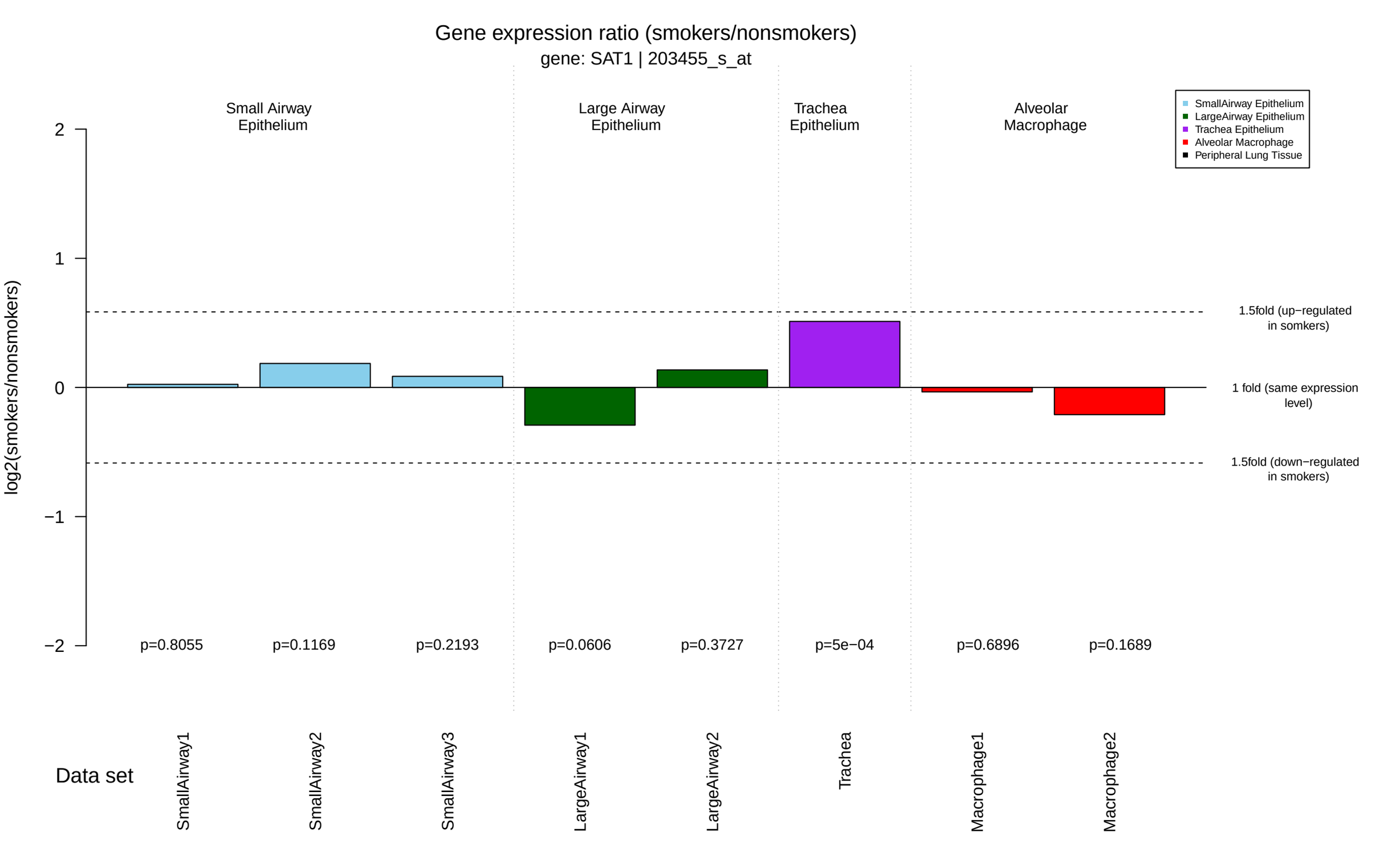
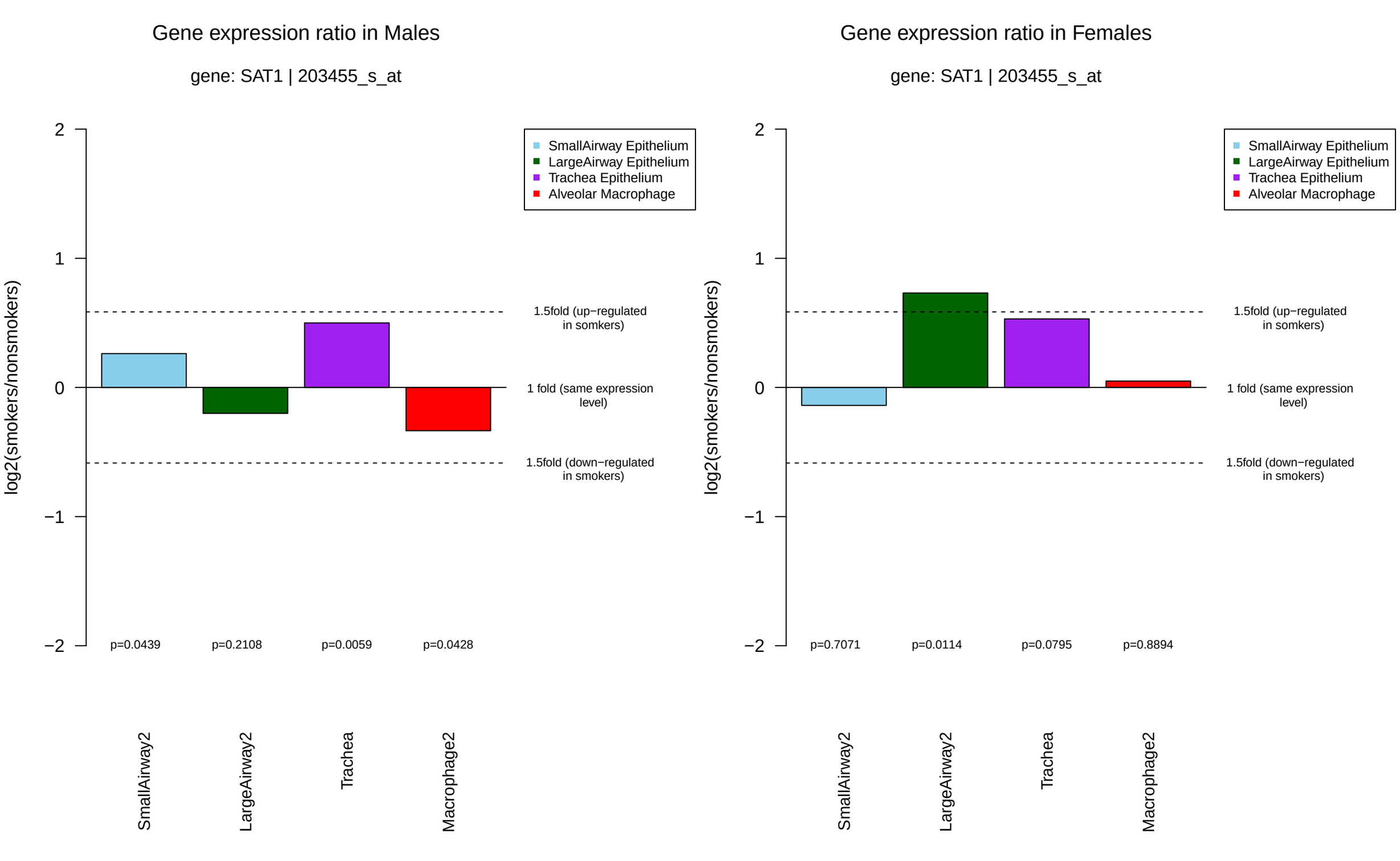
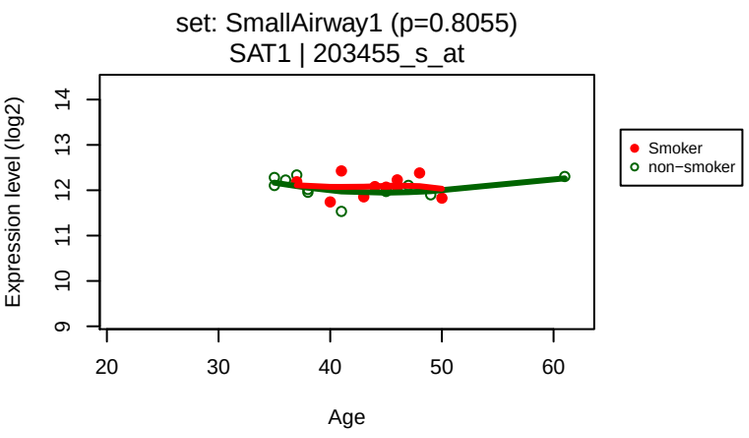
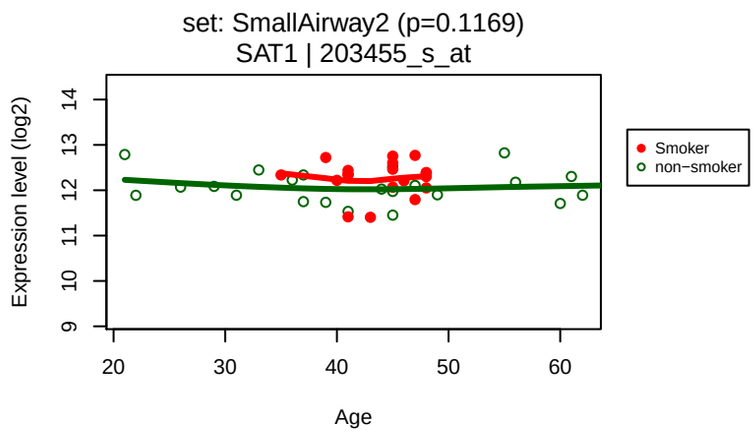
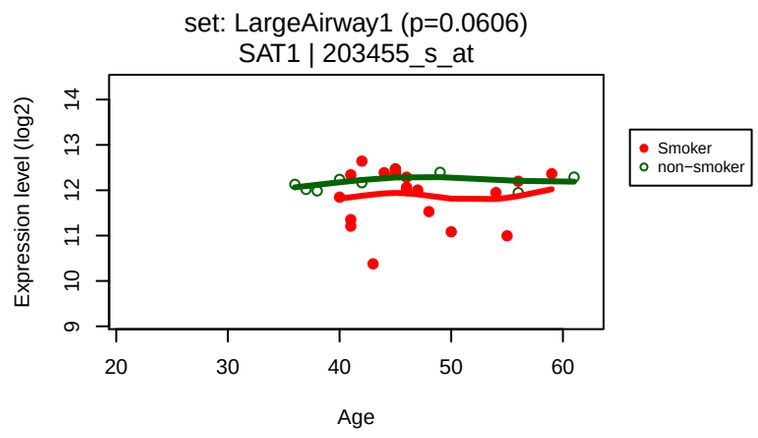

Gene expression of smokers and nonsmokers
gene: SAT1 | 203455_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8055
p=0.1169
p=0.2193
p=0.0606
p=0.3727
p=5e−04
p=0.6896
p=0.1689
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

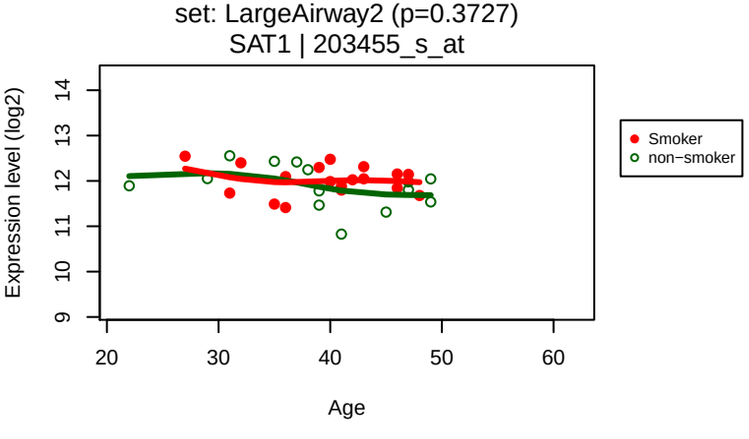

Gene expression ratio (smokers/nonsmokers)
gene: SAT1 | 203455_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8055
p=0.1169
p=0.2193
p=0.0606
p=0.3727
p=5e−04
p=0.6896
p=0.1689
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

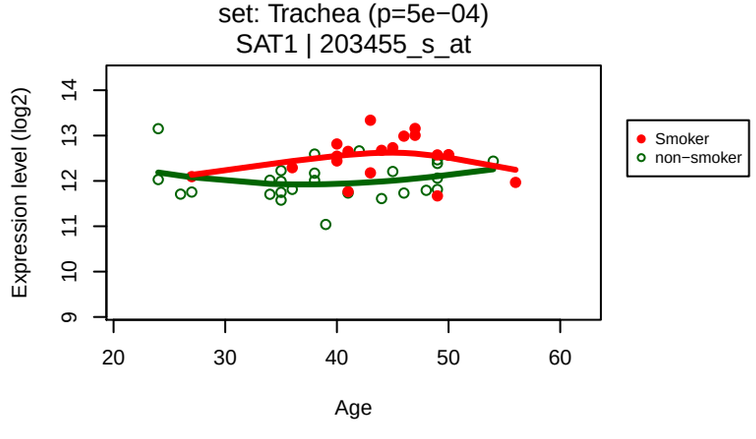

Gene expression ratio in Males
gene: SAT1 | 203455_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0439
p=0.2108
p=0.0059
p=0.0428
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SAT1 | 203455_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7071
p=0.0114
p=0.0795
p=0.8894
SmallAirway2
LargeAirway2
Trachea
Macrophage2

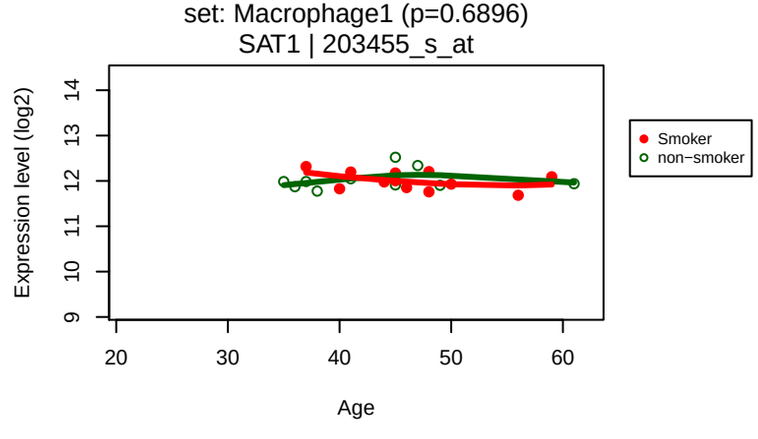

set: SmallAirway1 (p=0.8055)
SAT1 | 203455_s_at
Expression level (log2)
Age
Smoker
non−smoker

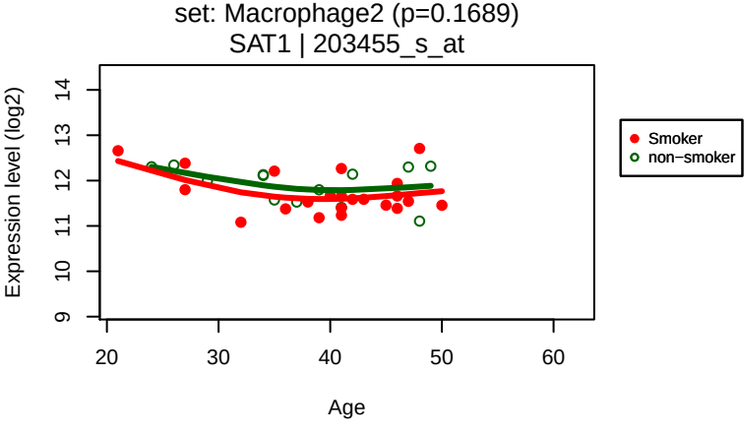

set: SmallAirway2 (p=0.1169)
SAT1 | 203455_s_at
Expression level (log2)
Age
Smoker
non−smoker

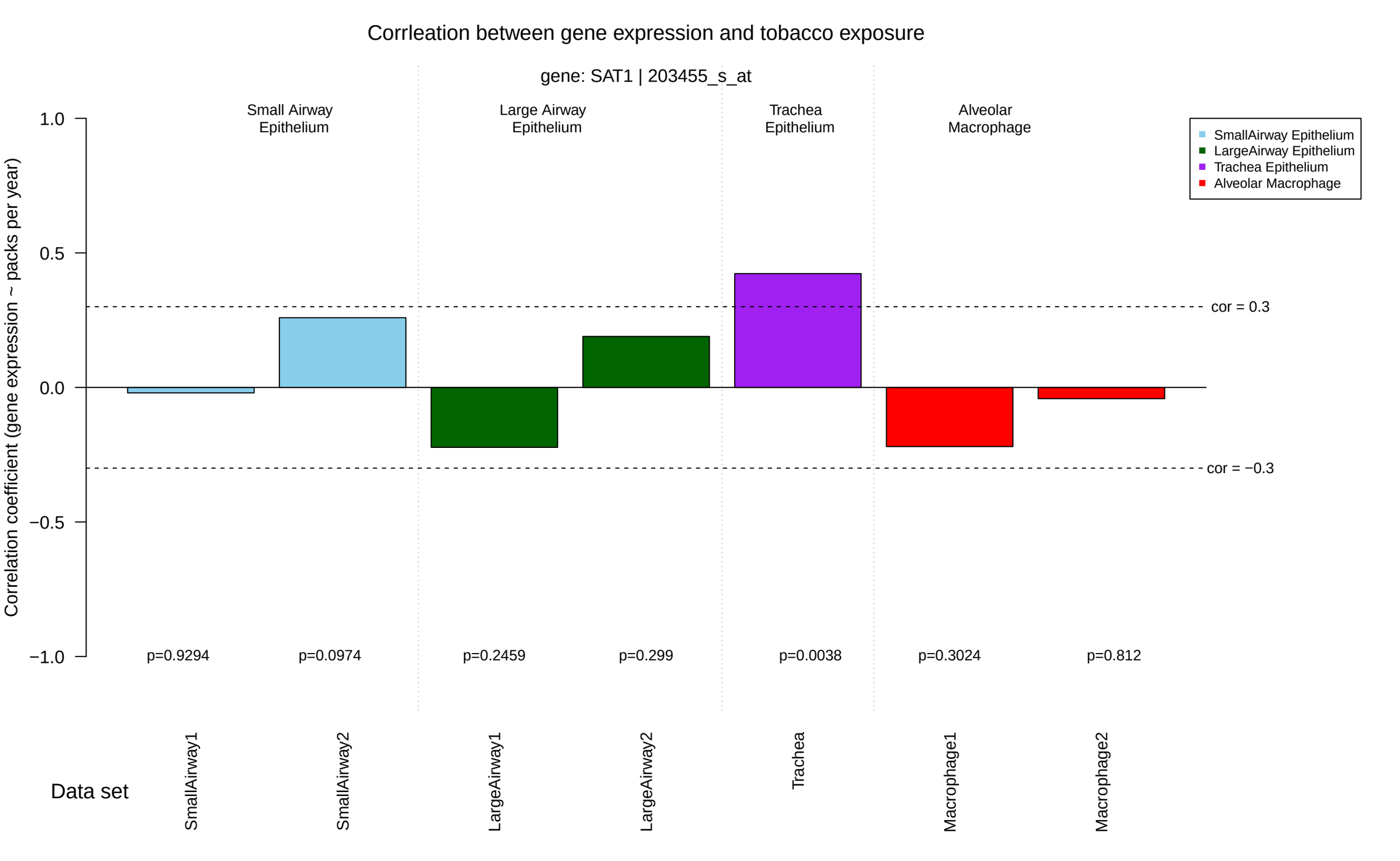

set: LargeAirway1 (p=0.0606)
SAT1 | 203455_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3727)
SAT1 | 203455_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=5e−04)
SAT1 | 203455_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6896)
SAT1 | 203455_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1689)
SAT1 | 203455_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SAT1 | 203455_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9294
p=0.0974
p=0.2459
p=0.299
p=0.0038
p=0.3024
p=0.812
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2